Biprakanta Pal

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Summary

Engineering leader with 10+ years across payments infrastructure, data platforms, and distributed systems at scale. Currently leading two engineering charters at Setu (a Pine Labs company) — BBPS bank-side payments (TSP for Axis, Canara, AU, Ujjivan, IDFC) and the core Platform group — after building and scaling the Identity/Data Gateway and Data Insights platforms from the ground up. Early and pragmatic adopter of AI-native engineering — AI-assisted development, code reviews, and reusable agent runbooks are embedded in how my teams ship. Proven record of taking platforms from 0→1, setting technical vision, running production incident command at bank-grade operational excellence bars, and building and managing high-retention engineering teams.

Experience

Tech Lead, Setu (a Pine Labs company) | *01/2023 – present* | *Bengaluru, India*

BBPS & Platform Engineering — Lead: Platform & BBPS, dual charter (2026–Present)

- Lead engineering for Setu's BBPS TSP business across bank operating units (Axis, Canara, AU, Ujjivan, IDFC, PhonePe-side OUs) — owning delivery cadence, standups, deployments, on-call, capacity planning for peak traffic events, and NPCI compliance including SSL signer certificate rotation across all BBPS stacks.
- Simultaneously lead the central Platform team (Kong API gateway, account service and OPA/OPAL authorization, Lagos billing, ClickHouse observability stack) serving every Setu product line.
- **Account-service crisis leadership:** ran incident command through recurring company-wide login outages — the account service is the auth backbone for every Setu product. Root-caused OPA/OPAL memory crashes to the OPAL server running on a single spot instance (reclaims forcing fleet-wide OPA sidecar reloads and OOM crashes); drove the structural fix — decoupling health and core auth APIs from OPA so logins survive OPA failure, and moving OPAL to redundant on-demand nodes — and owned internal and customer-facing RCA communications reviewed with CTO-level stakeholders.
- Drove RCA and remediation for a Kong/WAF rate-limit outage impacting Pine Labs aggregator and Plural traffic, including a WAF-rule review across platform and product layers.
- Diagnosed and same-day-fixed a billing charge-leakage incident during the Lago migration, and closed out preventive actions from prior RCAs including lag alerting on billing pipelines.
- Driving BBPS onboarding onto the Lagos billing platform (COU/BOU invoicing across OUs) — owning the technical roadmap and coordinating product, platform SPOCs, and data-pipeline owners across teams.
- Led new-OU delivery (AU Bank COU/BOU solutioning and UAT enablement, IDFC Clickpay 2.0) working directly with bank engineering counterparts.
- **Founded the BBPS Production Engineering (FDE) team:** authored the charter and JD, secured headcount, hired a Production Engineer Lead (SRE) and converted the first campus engineer — standing up bank-side operations: NPCI connectivity and certificate management, DR movement for bank OUs, and UAT/certification enablement.
- Own regulatory and network compliance for BBPS: RBI quarterly data submissions, NPCI MTLS migration readiness across all OUs, and DR drills coordinated with NPCI and partner banks.
- Cut infrastructure cost via ClickHouse log-retention policy adoption, S3 log reduction (targeting ~₹3L/month savings on top log producers), and RDS reserved-instance right-sizing.

SignalIQ / AutoDI — 0→1 product incubation (2026)

- Incubating AutoDI, a 0→1 bet under Setu's SignalIQ data products charter: automated bank-statement data ingestion via Account Aggregator and PDF flows feeding SignalIQ's underwriting analytics (BSA).
- Authored the AutoDI product overview and AA-flow integration guide, driving cross-functional review with product and data science; ensured Switchboard×SignalIQ billing integration was unblocked for commercial launch.

Data Gateway – Identity Products (2025–2026)

- Took over a mission-critical team post-org churn; rebuilt the team, stabilized operations, and drove availability from 99.5% (eSign) / 98.6% (RPD) toward a 99.9% SLO via proactive monitoring, telemetry and instrumentation investment, automated remediation, and a restructured dual-layer on-call model for operational excellence — zero DG-attributed P0 incidents across the cycle.
- **Switchboard (API Orchestrator):** owned the software architecture and shipped a scalable, vendor-agnostic orchestration platform end-to-end, applying cloud-native distributed-system design patterns — 3-layer routing (per-minute vendor health state machine with cooldown, six routing strategies incl. cost, latency, availability, success-rate and ratio-based, and preferred-partner priming) enabling fully automated partner failover with zero engineering intervention.

- Designed the BYOC (Bring Your Own Credentials) commercial model and SM/COST/CM billing convention; took Tata Capital live as the first BYOC customer paying a per-transaction orchestration fee with client-controlled routing.
- Migrated RPD fully onto Switchboard (CITI + Slice live) with a structured ramp plan — sandbox-first release, documented rollback, and QA gates; built automated AM-Lock failover that transparently reroutes Yesbank traffic to CITI.
- Shipped Bundled BAV (penny-drop + bank account verification) and eSign V4, delivering measurable conversion lifts for ETMoney (+10.4%), Univest (+5.6%), Angel One (+4.7%) and other marquee customers.
- Drove team delivery velocity and CI/CD automation: pod median PR merge time from 12 days → 1.2 days (10×), Docker layer caching for CI, and 6 reusable AI-assisted "Codex skill" runbooks that cut vendor-integration time from weeks to days; coached the team on disciplined AI-assisted development (3–4× faster cycles).
- Raised the quality and security bar: 60% unit/integration coverage, SonarQube quality gates enforced on every MR, caught and fixed a production PII-logging exposure, and owned three concurrent security audits (source code review, VAPT, data-localization SAR) with Cyraacs and PwC as platform owner.
- Crisis leadership: when a partner withdrew VPA-to-BAV access for a 60-RPS client (JAR), scoped a replacement product and coordinated build, QA, and billable metrics in parallel — shipped within hours of the issue surfacing; led 20+ epics across a 6-engineer team with proactive stakeholder communication.

Data Insights Platform (2023–2024)

- Spearheaded the Data Insights platform from POC to production, scaling throughput 15× (2 → 30+ TPS) via horizontally scaled async workers and prioritized ETL pipelines.
- Built configurable indicator pipelines and templatized transformer DAGs for dynamic ETL flows (<5s response); versioned data and models with DVC for consistent transformer development, debugging, and backfills.
- Established engineering best practices — Git strategy, gated CI/CD, automated testing, code-quality enforcement — and initiated the MLOps charter (Databricks, SageMaker, TrueFoundry evaluation).
- Led the architectural modernization decoupling transaction storage from processing; this platform foundation evolved into Setu's SignalIQ analytics suite.

Senior Software Engineer, Maximl Labs | *06/2021 – 12/2022*

- Designed an asynchronous Analytics Microservice (ServiceBus/SNS/SQS/NSQ) with CI/CD and Helm-based Kubernetes templating; built a cross-platform message-queue wrapper with unified APIs.
- Migrated a monolith to microservices; managed and mentored a team of 5 developers across chat support, audit logs, and integrations.

Senior Software Engineer, LeadSOC Pvt Ltd | *11/2020 – 05/2021*

- Built a data pipeline and visualization platform comparing performance metrics of chipsets across designs.

Software Engineer III, FactSet Research Systems | *06/2018 – 11/2020* | *Hyderabad*

- Developed multithreaded Python/C++ pipelines for global financial data ingestion; upgraded entitlement and research-document platforms; contributed to an RNN-based text predictor (88% accuracy).

System Engineer, Infosys Ltd | *03/2016 – 06/2018* | *Hyderabad*

- Built predictive monitoring and dynamic runbooks for semiconductor batch operations (LAM Research); centralized error recovery across 20,000 scripts via a searchable web app.

Technical Competencies

- **Payments & Fintech Platforms:** BBPS (COU/BOU/TSP), NPCI integrations, vendor orchestration & routing, billing/invoicing platforms (Lago), reconciliation & settlement, identity APIs (eSign, BAV, RPD, Aadhaar)
- **Languages & Backend:** Python, Golang, Java, Flask, FastAPI, Django, Spring Boot, Celery, gRPC, REST API design, Kong API Gateway, OPA/OPAL authorization
- **Data & Streaming:** PostgreSQL, Redis, Elasticsearch, ClickHouse, Kafka, Flink, Databricks, SageMaker, DVC, ETL/transformer DAGs, Account Aggregator data pipelines
- **Cloud & DevOps:** AWS (ECS, EC2, Lambda, S3, RDS, SNS/SQS), Azure, Kubernetes (Helm, Karpenter), Docker, Git, GitLab CI/CD, Terraform, SonarQube, Grafana/observability, cloud cost optimization
- **Delivery & Velocity:** streamlined delivery pipelines and CI optimization (Docker layer caching), PR cycle time 12d → 1.2d, AI-assisted development at 3–4× velocity, reusable engineering runbooks, release/rollback discipline
- **Reliability & Security:** Incident command & RCA, 99.9% SLO programs, on-call design, VAPT/SCR/audit ownership, PII safety

Leadership Highlights

- Dual-charter leadership: concurrently running BBPS payments engineering and the central Platform team — engineering, delivery, and reliability accountability across both.

- Trusted incident commander for company-wide P0s, driving cross-functional RCAs reviewed with CTO-level stakeholders and owning external customer communications.
- Team builder and people manager: rebuilt the Identity team from the ground up post-reorg, managing engineers directly — setting performance expectations, conducting performance evaluations, and running structured onboarding, knowledge-sharing, and AI-tooling coaching; low-attrition culture.
- **AI-native engineering champion:** instituted AI-assisted code reviews as early as 2023; drove team-wide AI-assisted development with practical guardrails (3–4× faster cycles, vendor integrations from 1–2 weeks to 1–2 days); published 6 reusable agent runbooks (Codex skills) standardizing how the team delegates implementation to AI; coach engineers on a plan-first, “treat-the-agent-as-an-intern” workflow that shifts developer effort toward solution design and AI code review.
- Own the engineering hiring charter: drove EM and SDE hiring across teams, founded and staffed the BBPS Production Engineering (FDE) function, and hired the founding member of the data science team.

Education

B.Tech, Mechanical Engineering — National Institute of Technology, Agartala *2015*

CGPA 7.36/10. Languages: English, Bengali, Hindi.